

chap_simples_BE

2025-07-09

PLAN BE

```
BE1 <- function(N,Pi)
{
  u <- runif(N, 0, 1)
  as.numeric(u<Pi)
}
```

```
set.seed(123456)
BE1(20,0.25)
```

```
## [1] 0 0 0 0 0 1 0 1 0 1 0 0 0 0 0 0 0 1 0 0
# [1] 0 0 0 0 0 1 0 1 0 1 0 0 0 0 0 0 0 1 0 0
```

```
set.seed(345678)
BE1(20,0.25)
```

```
## [1] 1 0 0 0 0 0 1 0 1 0 0 0 0 0 1 0 1 0 0 0
# [1] 1 0 0 0 0 0 1 0 1 0 0 0 0 0 1 0 1 0 0 0
```

```
BE2 <- function(N,Pi)
{
  n <- rbinom(1,N,Pi)
  srswor(n,N)
}
```

```
library(sampling)
set.seed(123456)
BE2(20,0.25)
```

```
## [1] 0 0 0 1 0 1 1 0 0 1 0 0 1 0 0 0 1 0 0 1
# [1] 0 1 0 1 0 0 1 1 1 0 0 0 0 0 0 1 0 1 0 0
```

```
set.seed(345678)
BE2(20,0.25)
```

```
## [1] 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 1
# [1] 0 0 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0 1 0 0
```

```
set.seed(123456)
pik <- rep(0.25,20)
UPpoisson(pik)
```

```
## [1] 0 0 0 0 0 1 0 1 0 1 0 0 0 0 0 0 0 1 0 0
# [1] 0 0 0 0 0 1 0 1 0 1 0 0 0 0 0 0 0 1 0 0
```

```
library(sampling)
data(rec99)
N <- nrow(rec99)
n <- 70
set.seed(123456)
s.be <- BE1(N,n/N)
sum(s.be==1)
```

```
## [1] 63
```

```
#63
```

```
rec.be <- rec99[s.be==1,]
(N/n)*sum(rec.be$LOGVAC)
```

```
## [1] 12156.34
```

```
#[1] 8761.114
```

```
(N/n-1)*(N/n)*sum(rec.be$LOGVAC^2)
```

```
## [1] 6133967
```

```
#[1] 2755398
```

```
pik <- rep(n/N,N)
s.be1 <- UPpoisson(pik)
sum(s.be1==1)
```

```
## [1] 74
```

```
#[1] 61
```